

Lecture 07

Hydrologic Forecasting

Review on Previous Lectures

- ❖ **Basic Probability Concepts:** Independence, Bayes' Theorem, ...
- ❖ **Probabilistic Modeling in Hydrology**
- ❖ **Flood Frequency Analysis**
 - ◆ Empirical Method (No Distributional Assumptions)
 - ◆ Analytical Method (Distributional Assumptions)
- ❖ **More Distributions for Extreme Events**
 - ◆ Generalized Extreme Value (GEV) Distributions
 - ◆ Peaks-over-threshold (POT) Approach (=Partial Duration)
 - Generalized Pareto Distribution (GPD)
 - ◆ How to Choose the Threshold u ? Mean Residual Life Plot

Today's Topic: Hydrological Forecasting

❖ Time Series Analysis

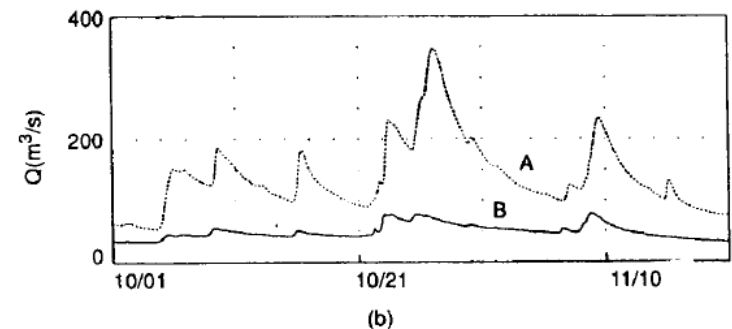
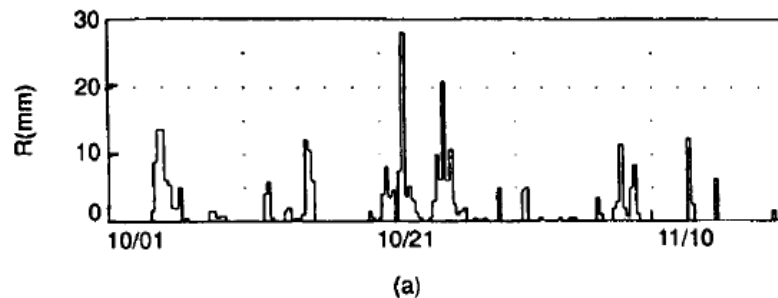
- ◆ Structure of Time Series Models -> Time Series Modeling
- ◆ Autoregressive Moving Average (ARMA) models
- ◆ Introduction to the "Hurst Phenomenon" (Long-term dependency)
- ◆ Why Important?
 - 시간에 흐름에 따라 발생하는 순차적인 데이터를 예측하는 대표적인 모수적 (parametric) 예측 방법이기 때문
 - 수문분야 외에도, 경제학, 주식, AI 등에서도 자주 사용되는 기법

❖ Ensemble Streamflow Prediction (ESP)

❖ Application in Research: SNU-CCAWR

Time Series Analysis: (1) Structure

- ❖ Single vs Multiple (Univariate vs. Multivariate)
 - ◆ One or Multiple hydrologic variable at a given site (or at multiple sites)
- ❖ **Uncorrelated vs. Correlated**
 - ◆ Independent vs. Dependent (Serially/Auto-correlated)
- ❖ Autocorrelation vs. Cross-correlation: Next Slide!
- ❖ Intermittent (a) vs. Non-intermittent (b)



Time Series Analysis: (1) Structure

❖ Autocorrelation vs. Cross-correlation

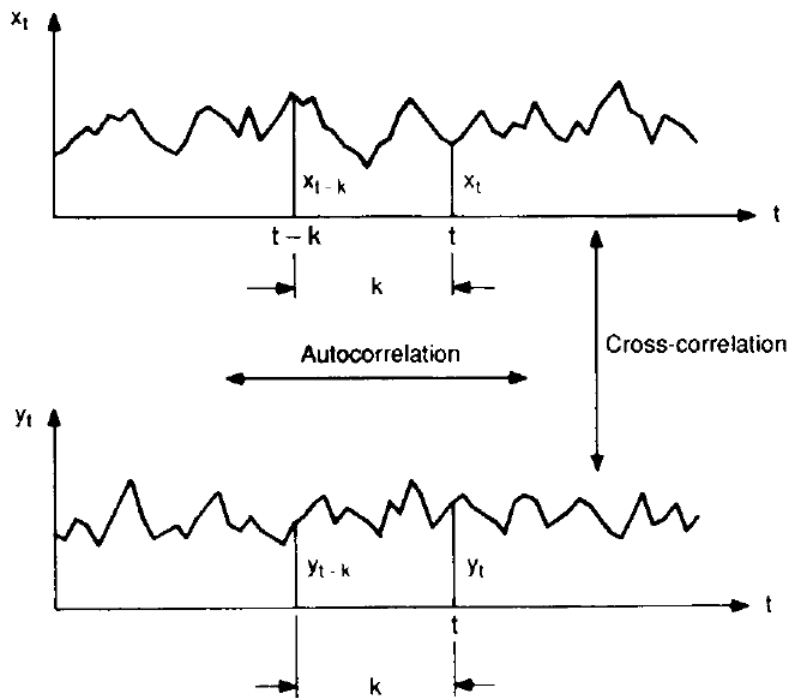


FIGURE 19.1.2 A pair of single time series x_t and y_t , each having autocorrelation. Cross-correlation is between the two series.

*Autocorrelation Coefficient (lag k)

The *sample autocorrelation function* r_k (or autocorrelation coefficients) of a time series may be estimated by⁸⁰

$$r_k = \frac{c_k}{c_0} \quad (19.2.4)$$

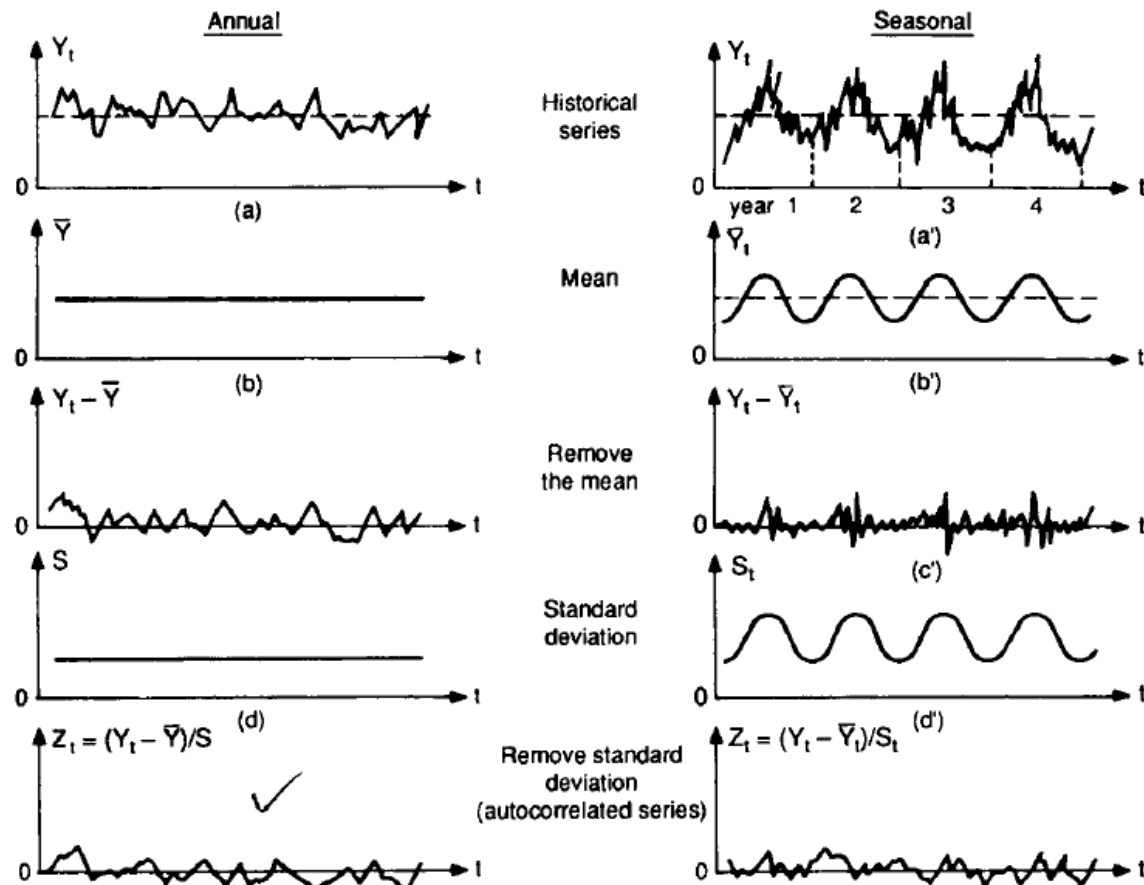
$$c_k = \left(\frac{1}{N} \right) \sum_{t=1}^{N-k} (y_{t+k} - \bar{y})(y_t - \bar{y}) \quad k \geq 0 \quad (19.2.5)$$

*Cross-correlation between X & Y (Correlation Coefficient)

$$\rho_{xy} = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

Time Series Analysis: (2) Partitioning

❖ Decomposition of Time Series



Time Series Analysis: (3) Modeling

❖ Autoregressive (AR) Models

AR Models. A time series defined as^{15,155}

$$y_t = \mu + \sum_{j=1}^p \phi_j (y_{t-j} - \mu) + \varepsilon_t \quad (19.3.1)$$

is called an *autoregressive model* of order p in which ε_t is an uncorrelated normal random variable (also referred to as *noise*, *innovation*, *error term*, or *series of shocks*) with mean zero and variance σ_ε^2 ; it is uncorrelated with $y_{t-1}, \dots, y_{t-p}$. Since ε_t is normally distributed, then also y_t is normal. The parameters of the model are $\mu, \phi_1, \dots, \phi_p$, and σ_ε^2 . The model, Eq. (19.3.1), is often denoted as the AR(p) model or simply the AR model. The AR(1) model takes the simple form

$$y_t = \mu + \phi_1 (y_{t-1} - \mu) + \varepsilon_t \quad (19.3.2)$$

Low-order AR models such as Eq. (19.3.2) have been widely used for modeling annual hydrologic time series (Refs. 42, 61, 114, 155, and 203, among others), and seasonal and daily series after seasonal standardization.^{138,207}

The mean, variance, and autocorrelation function of the AR(p) process are^{15,155}

$$E(y) = \mu \quad (19.3.3)$$

$$\text{Var}(y) = \sigma^2 = \frac{\sigma_\varepsilon^2}{\left(1 - \sum_{j=1}^p \phi_j \rho_j\right)} \quad (19.3.4)$$

$$\rho_k = \phi_1 \rho_{k-1} + \dots + \phi_p \rho_{k-p} \quad (19.3.5)$$

respectively. The last expression is known as the *Yule-Walker equation*. The three

Time Series Analysis: (3) Modeling

❖ Autoregressive Moving Average (ARMA) Models

ARMA Models. A more versatile model than the AR is the *autoregressive moving average model*¹²

$$y_t = \mu + \sum_{j=1}^p \phi_j (y_{t-j} - \mu) + \varepsilon_t - \sum_{j=1}^q \theta_j \varepsilon_{t-j} \quad (19.3.8)$$

with p autoregressive parameters $\phi_1, \dots, \phi_p$, and q moving average parameters $\theta_1, \dots, \theta_q$. The model is also denoted as an ARMA(p, q) model or simply as ARMA. Note that an ARMA($p, 0$) model is the same as an AR(p) model and the ARMA(0, q) model is the same as the *moving average model* MA(q). The noise ε_t in

Eq. (19.3.8) is an uncorrelated normal process with mean zero and variance σ_ε^2 and is also uncorrelated with $y_{t-1}, \dots, y_{t-p}$. A simple version of the ARMA(p, q) model is the ARMA(1, 1) as

$$y_t = \mu + \phi_1 (y_{t-1} - \mu) + \varepsilon_t - \theta_1 \varepsilon_{t-1} \quad (19.3.9)$$

Low-order ARMA models such as Eq. (19.3.9) are useful for operational hydrology in general, especially for modeling annual series^{23,67,68,107,115,129,155,156} and for seasonal series after seasonal standardization.^{33,116}

Hurst Phenomenon

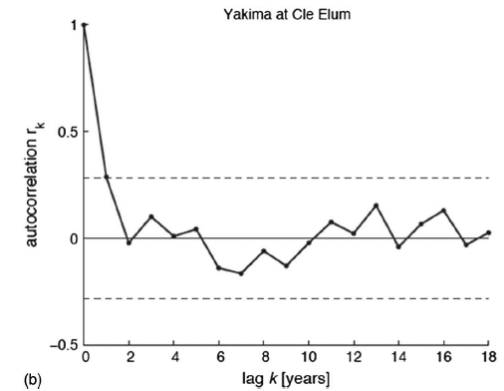
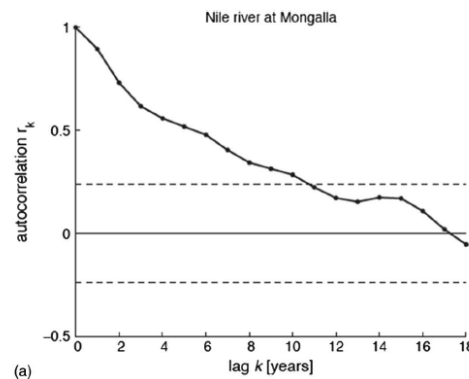
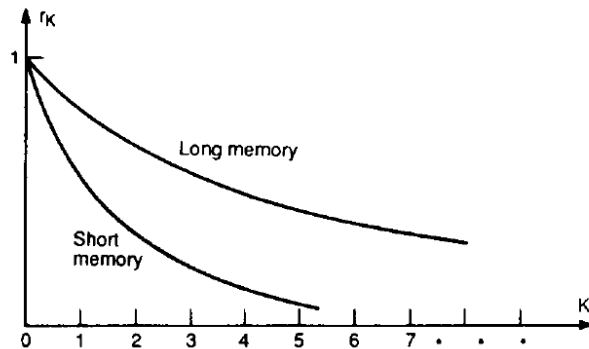
* Autocorrelation Coefficient (lag k)

The sample autocorrelation function r_k (or autocorrelation coefficients) of a time series may be estimated by⁸⁰

$$r_k = \frac{c_k}{c_0} \quad (19.2.4)$$

$$c_k = \left(\frac{1}{N} \right) \sum_{t=1}^{N-k} (y_{t+k} - \bar{y})(y_t - \bar{y}) \quad k \geq 0 \quad (19.2.5)$$

❖ Long vs. Short Memory



❖ Models for Long-term Variabilities

- ◆ Fractional Gaussian Noise (FGN)
- ◆ Long Memory Models: FARMA Model

Fractionally differenced Autoregressive Integrated Moving Average Models

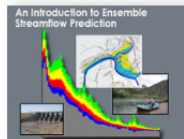
Ensemble Streamflow Prediction (ESP)



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📖 An Introduction to Ensemble Streamflow Prediction +

Available online: https://www.meted.ucar.edu/training_module.php?id=246



Published by: The University Corporation for Atmospheric Research ; 2007

The “Introduction to Ensemble Streamflow Prediction” module provides basic information on probabilistic streamflow forecasting. In this webcast, Dr. Richard Koehler, the National Hydrologic Sciences Training Coordinator for NOAA's NWS, presents information about the types of organizations that might use probabilistic streamflow forecasts as well as foundation concepts and background for ESP methods. The module begins with a brief review of hydrologic models including deterministic, stochastic, and scenario-based approaches. It then provides an overview of time-series approaches including a summary of traditional techniques such as flood frequency, flood analysis, statistical analysis, and trend analysis. Finally, the module presents the basics of ESP techniques including an explanation of its strengths, weaknesses, and appropriate application. The module also provides guidance on

❖ Source:

https://library.wmo.int/index.php?lvl=notice_display&id=21098#Y4gCenbP2UI

ESP: Introduction

❖ Hydrologic Models: Simulate & forecast Streamflow

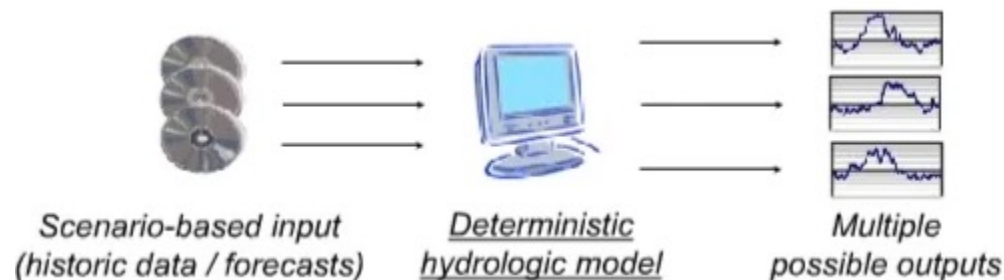
◆ Types of Hydrologic Models

- Accounting-based: Budget approach
- Physics, Empirical, Statistical-based Models

◆ Always results in **Model Error & Forecast Uncertainty**

◆ Classified as either **deterministic or stochastic (probabilistic)**

- Deterministic: Always same result for a given input
- Stochastic: Multiple possible outcomes (ex. Markov Chain models)
- Scenario-based deterministic model



Ensemble Streamflow Prediction

❖ ESP: a “scenario-based” approach

- ◆ Captures current model state conditions (hydrologic states)
- ◆ Input: Historic or forecast climate data (ex. Daily precipitation, Minimum/maximum daily air temperature, wind speed, etc.)
- ◆ Output: future streamflow & related probabilities (ex. Storage, discharge, runoff, exceedance probability, likelihoods, etc.)

◆ **NOT a stochastic model**

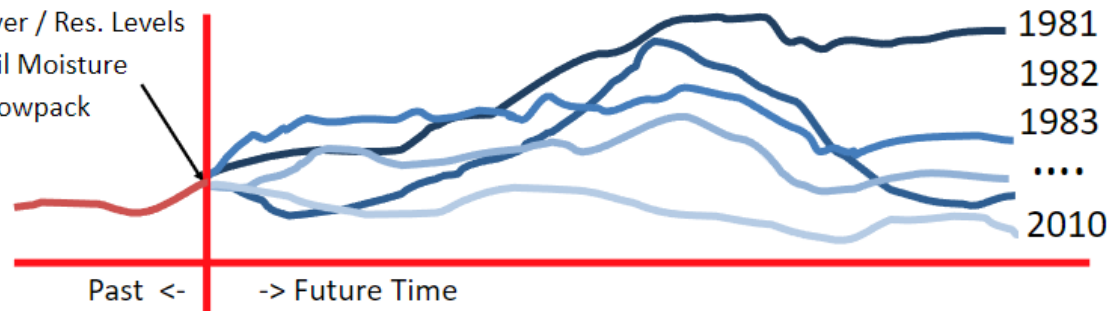
◆ Only model forcing uncertainty is captured in the probabilities

Current hydrologic states :

River / Res. Levels

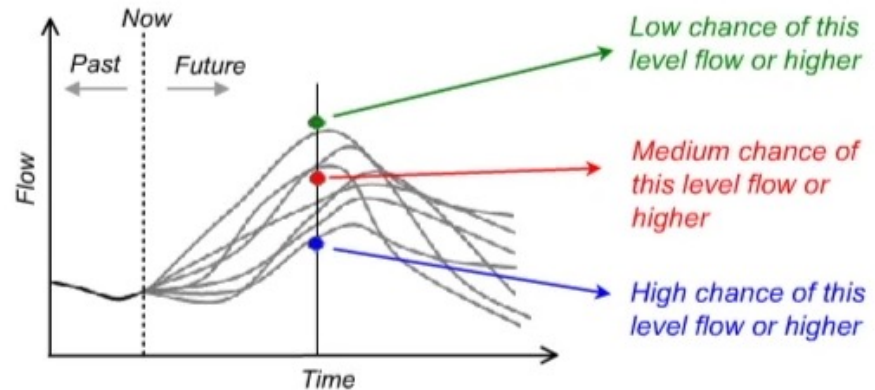
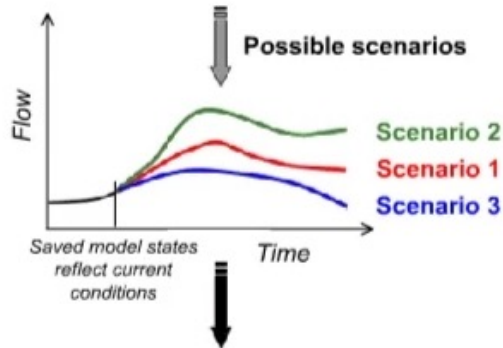
Soil Moisture

Snowpack



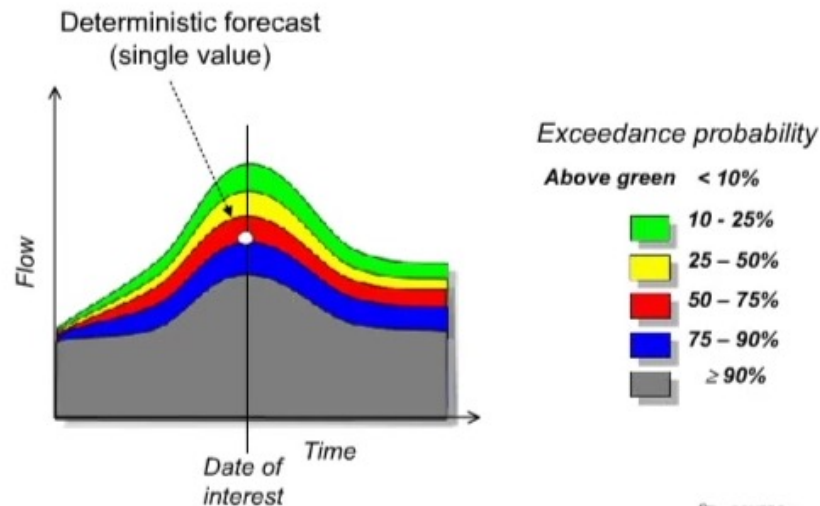
ESP Techniques

Multiple streamflow scenarios with historic meteorological or forecast weather/climatic data



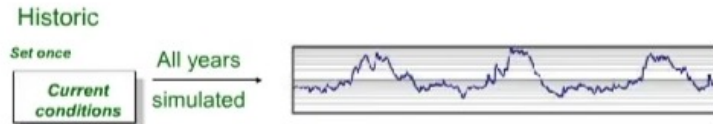
Results used in statistical analysis to produce forecasts with probabilistic values

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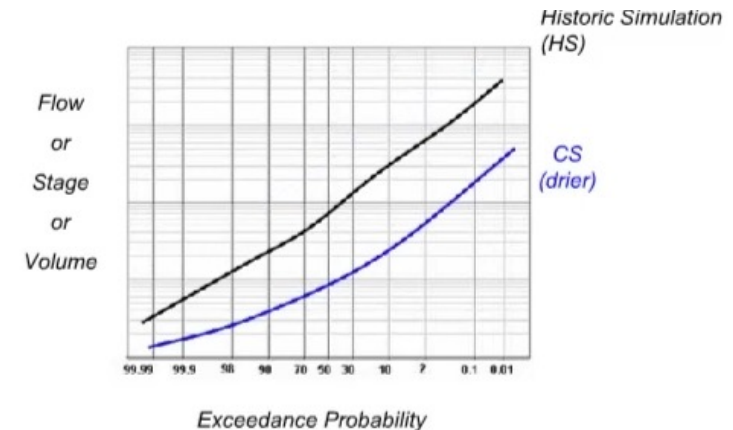
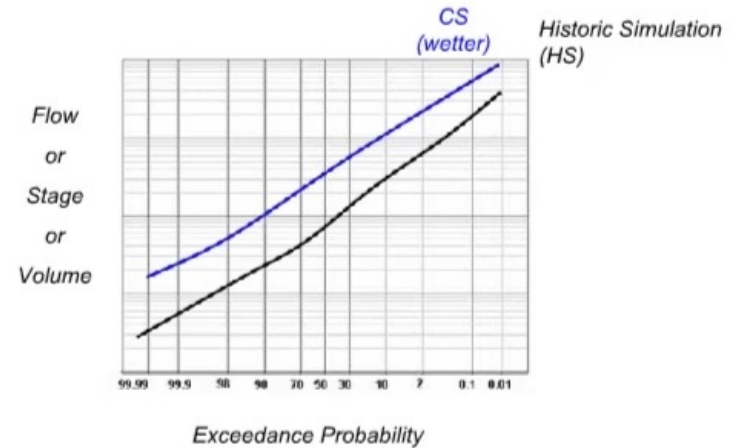
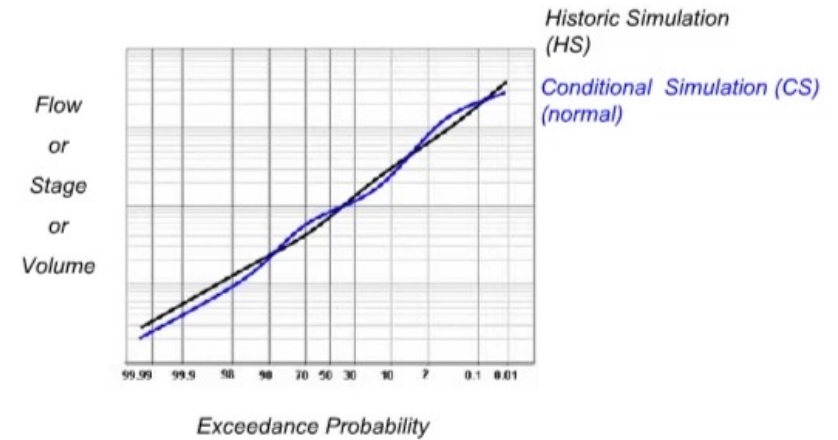
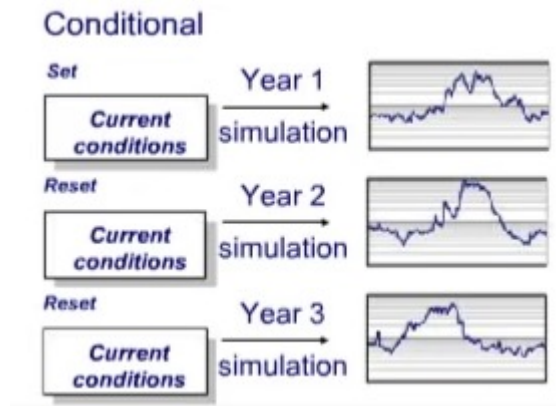


ESP Simulation

❖ Historical simulation

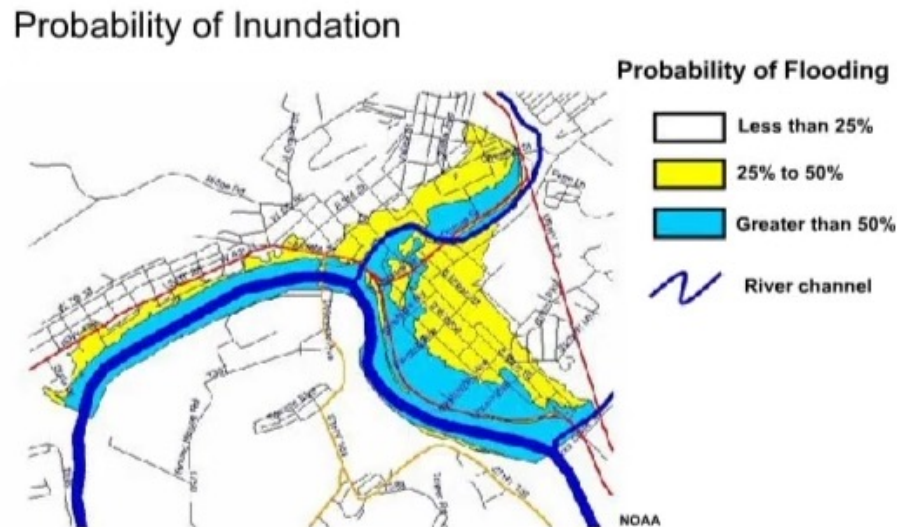


❖ Conditional simulation



ESP Results (Probabilistic Analysis)

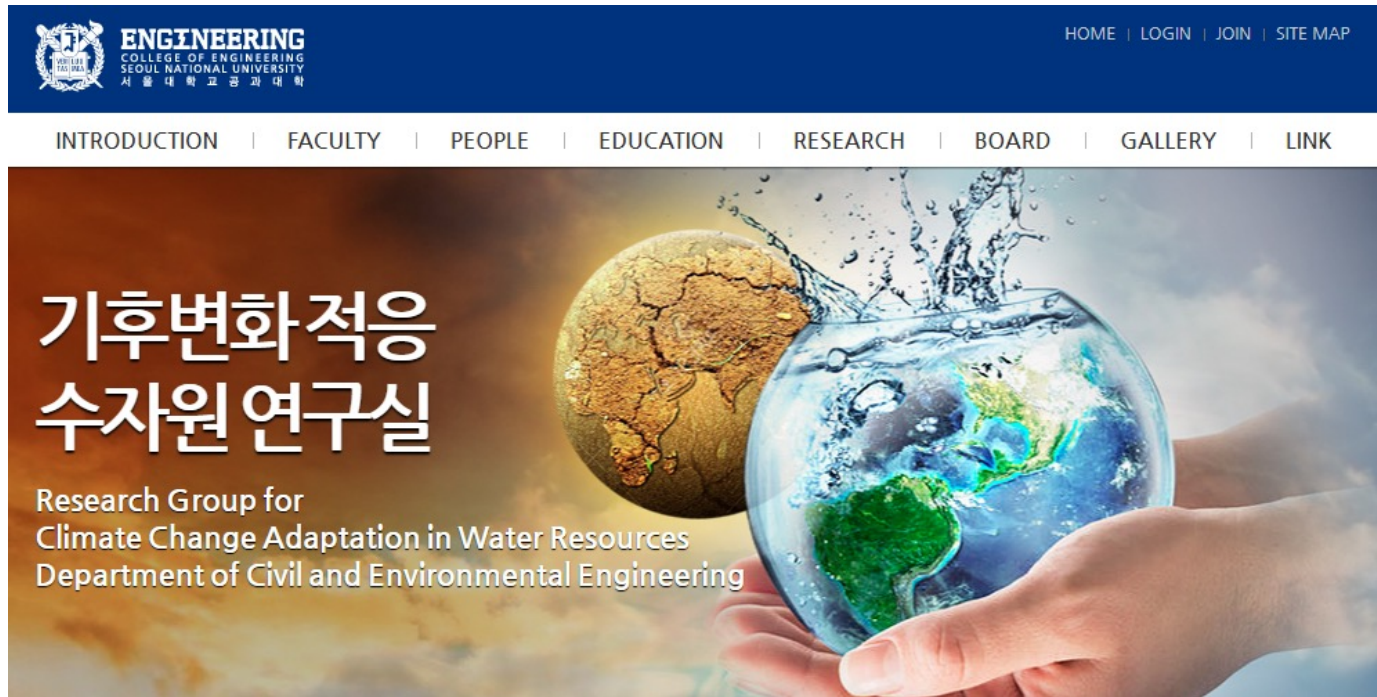
- ❖ ESP Products for probabilistic analysis include...
 - ◆ Trace plots, Expected value plot, Probability plot, Exceedance plot
 - ◆ Probabilistic Flood Mapping (with GIS tools)



Applications in Research

- ❖ Seoul National University-CCAWR Research Group
(Climate Change Adaptation in Water Resources)

<http://ccawr.snu.ac.kr/>



Future Schedule Preview

❖ 다음 주 강의 (2022-12-13 화요일 18:30-21:20)

◆ ***Final Project Presentation!***

❖ Future Schedule (~End of Semester)

◆ **12/13: Final Project Presentation (Presentation Due)**

◆ **12/20: Final Project Report Due (By Email)**